Dynamic EMS Dispatch & Station Placement Optimization

Integrated station siting and ambulance dispatch on **2.2 million** real Seattle 911 records

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Operations Research | Final Review

Survival is decided in minutes

Two operational decisions control how fast an ambulance arrives

Q1 — Where should ambulances wait?

Place stations well and more of the city sits inside a short response radius. A *strategic* decision, made once.

Set covering / maximal covering — ILP

Q2 — Who should be sent?

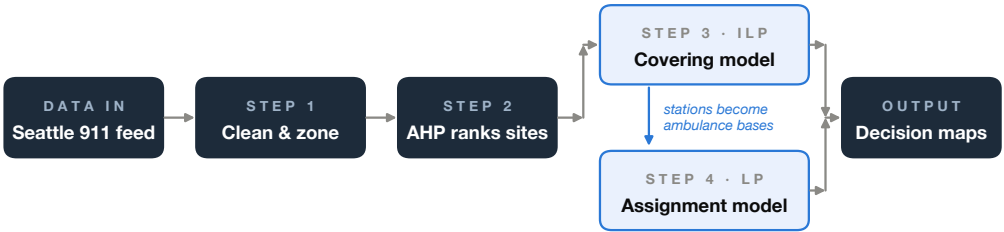
When several emergencies land at once, the right ambulance-to-incident matching cuts everyone's wait. An *operational* decision, made every hour.

Assignment problem — LP

The two are linked: the stations chosen by the covering model *become* the ambulance bases the assignment model dispatches from.

The pipeline

Five stages, each one feeding the next



Every stage runs on the real feed. No synthetic demand, no sampled subset, no hand-picked study area.

All three models solve locally in milliseconds — the whole pipeline is interactive, not a batch job.

The data

Three public sources, no API keys, fully reproducible

Source	What it is	Used for
Seattle Real-Time Fire 911 Calls	2,197,739 records, 2003–2026	Where and when emergencies occur
Seattle Fire Stations (City GIS)	The 36 stations Seattle runs	Benchmarking our placement
ZIP Code Tabulation Areas (Census)	ZIP polygons	Assigning each incident a zone

79,152 EMS incidents
study year, 27 Aug 2025 – 27 Aug 2026

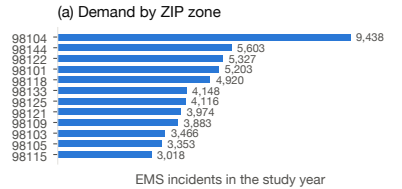
28 demand zones
after a materiality floor

36 real stations
the benchmark network

The raw feed mixes fire, alarms and hazmat with EMS; calls are filtered to genuine aid/medic dispatches by whole-word patterns, de-duplicated on incident number, and tagged by point-in-polygon join.

Step 1 — Demand is concentrated, not uniform

Which is exactly why placement can beat history



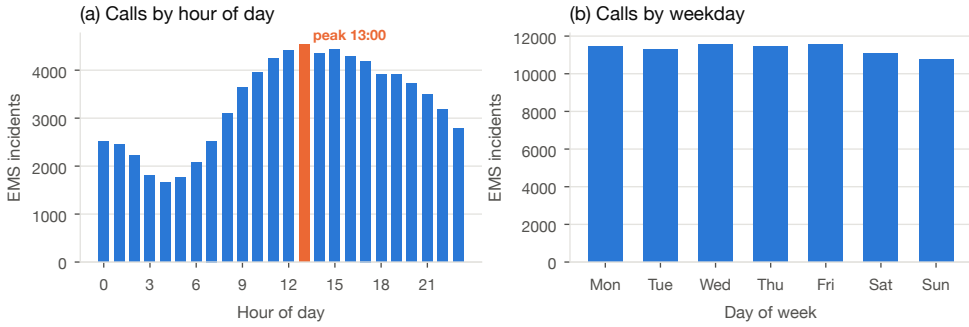
The 15 busiest ZIP zones carry **80.0%** of all 79,152 incidents.

Zone 98104 alone accounts for **9,438** calls — 11.9% of the city's EMS demand in one ZIP.

Each zone becomes a demand point weighted by its incident count, located at the *mean incident coordinate* rather than the polygon centroid — the candidate site sits where the calls actually are.

Step 1 — When the load spikes

A midday peak, and a flat week



Demand peaks at **13:00** and runs roughly **216 calls/day**. The weekday spread is only a few percent (Fri 11,577 against Sun 10,757), so the *hour* matters for dispatch while the *day* does not — which is why the dispatch scenario is chosen by peak hour, not by weekday.

Step 2 — AHP: turning three competing criteria into one score

Criterion	Weight
Incident Density	**67.4%**
Equity Gap	23.1%
Coverage Potential	9.4%

Four members rate the three criteria pairwise on Saaty's 1–9 scale, independently. Matrices are combined by **element-wise geometric mean** — the only average that keeps the matrix reciprocal.

Weights are the principal eigenvector. Human judgements can quietly contradict each other, so AHP measures that:

$$\mathrm{CI} = \frac{\lambda_{\max} - n}{n - 1}, \qquad \mathrm{CR} = \frac{\mathrm{CI}}{\mathrm{RI}}$$

$\lambda_{\max}$	CI	RI	CR
3.0000	0.0000	0.58	**0.0000**

$\mathrm{CR} < 0.10$ for all four members *individually* and for the aggregate.

CR is exactly zero by construction, not by luck: each member's density-vs-coverage rating is the product of their other two, so the judgements contain no contradiction for the check to find.

Step 3 — Optimisation 1: where to put the stations

Mode A — Set covering

Fewest stations that put *every* zone in reach.

$$\begin{aligned} \min\ & \textstyle\sum_i y_i \\ \text{s.t. } & \textstyle\sum_{i \in N(j)} y_i \ \geq\ 1 \quad \forall j \\ & y_i \in \{0,1\} \end{aligned}$$

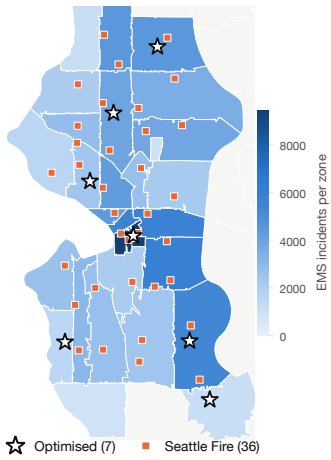
Mode B — Maximal covering

On a fixed budget P, cover the most *demand*.

$$\begin{aligned} \max\ & \textstyle\sum_j d_j z_j \\ \text{s.t. } & z_j \ \leq\ \textstyle\sum_{i \in N(j)} y_i \quad \forall j \\ & \textstyle\sum_i y_i = P, \quad y_i, z_j \in \{0,1\} \end{aligned}$$

$N(j)$ is the set of candidate sites that reach zone j inside the limit; d_j is the zone's incident count, so Mode B maximises *people reached*, not area or zone count. Solved exactly with HiGHS via PuLP — **5 ms** on the study instance.

Step 3 — The eight-minute solution



7 stations

100.0% of demand covered

28/28 zones inside 8 minutes

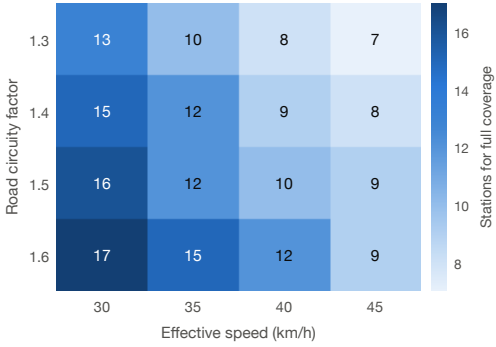
5 ms to prove optimality

Against Seattle's **36** real stations at the same limit.

But hold that number — the next slide is why *7* should never be quoted on its own.

How much of this is Seattle, and how much is our assumption?

The slide we would want a reviewer to ask for



The 911 feed records **no response times**. Every minute in this project is computed:

$$t = \frac{d_{\text{haversine}} \times \text{circuity}}{\text{speed}} + \text{turnout}$$

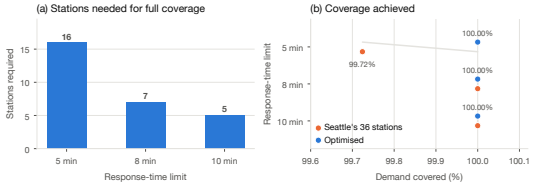
So we re-solved the ILP across a 4×4 grid of those two constants — 16 exact solves.

7–17 stations, a 2.4× spread

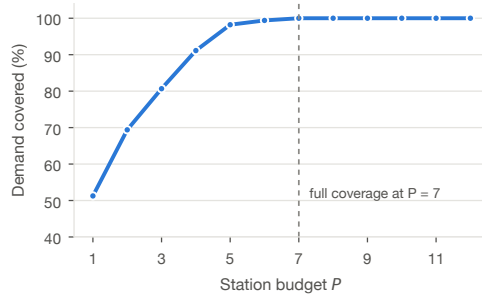
100% coverage in *all* 16 cells

The count is assumption-dependent. Being under 36 is not.

Step 3 — What a tighter promise costs



Stations needed per response limit



Diminishing returns against budget P

Tightening the promise from 10 to 5 minutes takes the network from **5** stations to **16** — the cost of a guarantee is convex. The budget curve is the argument for Mode B: the first handful of stations buy most of the city, and the last few buy very little.

Step 3 — Against the network Seattle actually runs

Response limit	Our stations	Demand covered	Seattle's 36
5 min	16	100%	99.72%
8 min	7	100%	100%
10 min	5	100%	100%

What this supports

Placement that follows *measured demand* reaches the whole city from fewer sites than placement accumulated historically. Directionally true in every cell of the robustness grid.

What it does not

The 36 are *fire* stations — they also carry suppression, hazmat and technical rescue, missions this model does not place for. The comparison is not like-for-like, and the ratio flatters us.

Step 4 — Optimisation 2: who to send

$$\begin{aligned} \min \ & \textstyle\sum_i \sum_j c_{ij}\, x_{ij} \\ \text{s.t.} \ & \textstyle\sum_i x_{ij} = 1 \quad \forall j \\ & \textstyle\sum_j x_{ij} \le 1 \quad \forall i \\ & x_{ij} \ge 0 \end{aligned}$$

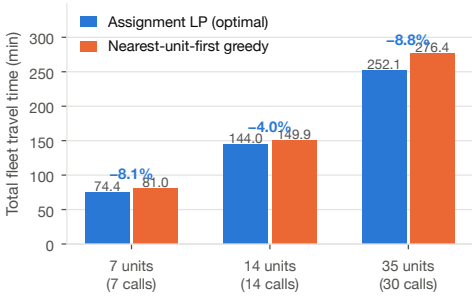
c_{ij} is travel time from ambulance i to incident j. The constraint matrix is totally unimodular, so the LP relaxation is integral — no branching needed.

Scenario: the busiest hour in the study year — Wed 19 Nov 2025, 09:00, **30 simultaneous incidents**, dispatched from the **7 stations the covering model opened**.

A rectangular cost matrix is padded with zero-cost dummy rows or columns so the problem is square and balanced; dummies never distort the real total.

Cross-checked against an independent Hungarian-algorithm solution: identical to six decimal places.

Step 4 — Optimal dispatch against nearest-unit-first



Fleet	Optimal	Greedy	Gain
35 units *(default)*	252.1	276.4	**8.8%**
14 units	144.0	149.9	4.0%
7 units	74.4	81.0	8.1%

Both sides clear the same calls. A peak hour has more calls than units, and unserved calls cost nothing in the objective — so an unrestricted LP would quietly answer only the *nearest* calls while greedy took them as they came.

Comparing those totals would score *triage* rather than *routing* and inflate our margin to 62%.

What is measured, and what is modelled

Stated before it is asked

Measured

- Incident locations, times and counts — 2.2M actual records
- The 36 real station locations
- ZIP zone geometry
- Every optimisation result, solved exactly and verified

Modelled

- **Every response time** — the feed contains none
- Demand aggregated to 28 zone centroids
- AHP weights — subjective by design, hence the CR test
- Units assumed always free (no queueing)

This is an optimisation over **real demand geography** with a **modelled travel layer**. Its claims are therefore comparative — which sites, in what order, how the answer moves — rather than predictive. Which is why the robustness grid is a result slide and not an appendix.

Verification — the models are not taken on trust

- Set covering matched **brute force** on 25 fully enumerated instances
- Maximal covering matched brute force on 25 instances
- AHP weights matched the largest-modulus eigenvector on **200 random matrices**
- Optimal dispatch never lost to greedy over **150 random cost matrices**
- Assignment LP matched an independent **Hungarian** solution exactly

64 model checks

31 application tests

0 failures

Including monotonicity of the robustness grid: a slower fleet or twistier roads can never need *fewer* stations — which is what separates a real gradient from solver noise.

It is a live application, not a script

Four tabs, every model re-solving interactively:

- **Data** — demand in space and time
- **Site scoring** — AHP, live Saaty sliders
- **Station placement** — both ILP modes, sensitivity, robustness, benchmark
- **Dispatch** — the LP on any peak hour

Every assumption is a control, not a constant:

- Study period, trauma inclusion
- Response-time limit
- Speed, circuity, turnout
- Station budget, fleet size

Deployed on Streamlit Community Cloud; solves run in-process, so nothing leaves the container.

Conclusions

- Demand-driven placement reaches the whole city from **fewer sites** than Seattle's historically accumulated network — the finding is directional and survives every travel assumption we tested.
- The station count itself is **7–17**, not 7. Reporting the range is the result; reporting the point estimate would have been the mistake.
- Optimal dispatch beats nearest-unit-first by **8.8%** of total fleet travel time on the busiest hour, measured like-for-like.
- The two models **compose**: covering output becomes assignment input, so a siting decision is evaluated by the dispatch performance it enables.

Next: replace the modelled travel layer with real road-network distances, and extend the covering model to **MEXCLP** (Daskin, 1983) so unit busy-fraction is priced in rather than assumed away.

Thank you

Questions

2.2M real records | 3 OR models | 95 automated checks, 0 failures